MIL-T-23142B
18 August 1977
Superseding
MIL-T-23142A
15 February 1966

MILITARY SPECIFICATION

# TAPE, PRESSURE-SENSITIVE ADHESIVE, FOR DISSIMILAR METAL SEPARATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for one type of pressure-sensitive tape for separation of dissimilar metals. (See 6.1.)

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| QQ-A-250/5 | Aluminum Alloy, Alclad 2024, Plate and Sheet |
| QQ-A-250/13 | Aluminum Alloy, Alclad 7075, Plate and Sheet |
| QQ-M-44 | Magnesium Alloy, Plate and Sheet (AZ31B) |
| TT-S-735 | Standard Test Fluids, Hydrocarbon |
| TT-T-291 | Thinner, Paint, Volatile Spirits (Petroleum Spirits) |
| UU-P-31 | Paper, General Specifications and Methods of Testing |
| PPP-T-680 | Tape, Pressure-Sensitive Adhesive, Packaging and Packing of |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7510

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS (Continued)

Military

| | |
|---|---|
| MIL-M-3171 | Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on |
| MIL-C-8514 | Coating Compound, Metal Pretreatment, Resin-Acid (ASG) |
| MIL-P-23377 | Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant |
| MIL-C-81773 | Coating, Polyurethane, Aliphatic, Weather Resistant |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-191 | Textile Test Methods |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

* 3.1 First article inspection. The tapes furnished under this specification shall be a product which has been inspected and has passed first article inspection herein (see 4.3).

3.2 Materials. The materials used in the manufacture of tape pressure-sensitive adhesive for dissimilar metal separation shall be such as to ensure performance of the tape in conformance with the requirements of this specification.

3.2.1 Adhesive. The adhesive shall be pressure-sensitive, water insoluble, homogeneous, and coated in a smooth and evenly distributed layer on one side of the backing. The adhesive shall require no solvent, heat, or any other preparation prior to application.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

3.3 Finished tape.

3.3.1 Rolls. The tape shall be uniformly and smoothly wound in rolls, with the adhesive side in, on suitable cores having an inside diameter of 3 +1/16, -0 inches. The core shall have sufficient rigidity to prevent distortion of the roll under normal conditions of use. The tape shall be in one continuous strip, except that any single roll may contain three splices provided the percentage of spliced rolls in any shipment does not exceed the manufacturing practice.

3.3.2 Width and length. The width of the tape shall be 1 inch unless otherwise specified in the contract or order (see 6.2). The length of tape in the rolls shall be a minimum of 72 yards or as specified in the contract or order (see 6.2).

3.3.3 Marking of rolls. Each roll of tape shall be marked on or near the edge of the core with numerals or letters indicating the month and year of manufacture. In addition, the manufacturer's name and designation of the product and the specification number shall be stamped in the core.

3.4 Physical properties. When tested as specified in Section 4, the tape shall conform to the requirements specified in Table I.

3.5 Workmanship. The process of the manufacture shall be such as to assure compliance with the requirements of this specification. The adhesive shall be applied to the backing in a smooth and uniform coating, and the finished tape shall contain no bare spots or lumps.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection of the tape shall be classified as follows:

a. First article inspection (4.3).

b. Quality conformance inspection (4.4).



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

TABLE I. Test Requirements.

| Property | Unit | Requirement | Test Paragraph |
|---|---|---|---|
| Thickness | inch (max) | 0.004 | 4.7.1 |
| Breaking Strength | lbs/inch width (min) | 15 | 4.7.2 |
| Adhesion | | | |
| To magnesium | ozs/inch width (min) | 20 | 4.7.3.2.1 |
| After accelerated aging | ozs/inch width (min) | 14 | 4.7.3.2.2 |
| After exposure in petroleum base oil | ozs/inch width (min) | 14 (with 1/8 inch edge lifting permitted) | 4.7.3.2.3 |
| After exposure in di-ester fluid | ozs/inch width (min) | 14 (with 1/8 inch edge lifting permitted) | 4.7.3.2.3 |
| Effect of Paint | - - - | No lifting, no shrinkage | 4.7.4 |
| Corrosion | - - - | No corrosion, pitting, or etching on magnesium surface covered by tape | 4.7.5 |
| Weathering (Accelerated outdoor exposure) | - - - | After one year outdoor exposure, no corrosion, pitting or etching on magnesium surface covered by tape. No lifting of 1/4 inch exposed area permitted. No delamination permitted. | 4.7.6 |
| Accelerated aging | - - - | No adhesive flow on sides, no telescoping, no distortion greater than 1/2 inch. | 4.7.7 |
| Shelf-storage | - - - | After shelf-storage, there shall be no adhesive flow on sides, no telescoping, no distortion greater than 1/2 inch and shall meet the Adhesion to magnesium requirement. | 4.7.8 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

* 4.3 First article inspection. First article inspection shall include all the examinations and tests of this specification except the examinations of preparation for delivery. Unless otherwise specified, no tape shall be submitted for acceptance under any contract or order until the first article samples prescribed in 4.3.1 have been subjected to first article inspection and pronounced satisfactory by the procuring activity. However, approval of the first article samples shall not relieve the supplier of his obligation to meet the quality conformance inspection (4.4). The responsibility for the performance of the first article inspection shall be specified by the procuring activity (see 6.3.1).

* 4.3.1 First article test samples. Samples for first article inspection shall consist of at least 5 rolls of 1-inch tape. Samples shall be selected at random from material which has been manufactured using the same production processes, procedures and equipment which will be used in filling the contract.

* 4.3.1.1 Data to accompany test samples. Two copies of the manufacturer's test report shall be submitted simultaneously with the samples of 4.3.1. The report shall contain numerical test data, where applicable, showing that the tapes submitted for the first article inspection conform to the requirements of this specification and a brief description of the components or raw materials used in the manufacture of the tape. Information shall be furnished, including plant address(es), as to the plant(s) in which the tape is or will be manufactured. If more than one plant address is listed, a certificate of equivalence of other plants to the plant in which the sample is manufactured must be furnished. The first article samples and test data shall be submitted as designated by the procuring activity.

* 4.3.2 First article sample and inspection for a subsequent contract. If a contractor has previously furnished tapes in accordance with the requirements of this specification and his product has been found to be satisfactory, the requirements for a first article sample and its submittal for any subsequent contract or order may be waived at the discretion of the procuring activity (see 6.3.1).

4.3.3 First article tests. First article tests are those tests accomplished on samples submitted to first article as a satisfactory product. The tests of the tapes shall consist of all the tests of this specification except for a certificate of compliance on weathering (accelerated outdoor exposure) test and the one year shelf-storage tests. Acceptance will be granted only upon satisfactory completion of all required tests and the submitting of the required certifications.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

* 4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations specified in 4.5 and the tests listed in Table II for each lot of tapes submitted for acceptance.

* 4.4.1 Inspection lot. A lot shall consist of tape of the same type and thickness, manufactured by a process that is essentially unchanged within a two week period, with components each of which have been produced at one time, at one plant, by one manufacturer. Lot size shall be defined as the number of such rolls produced in a two week period but shall not exceed 10,000 rolls.

* 4.4.2 Quality conformance test samples. A random sample of tape shall be taken from each inspection lot in accordance with inspection level S-1 of MIL-STD-105, where the unit of product shall be 5 rolls of 1-inch tape and the acceptance number shall be zero. The sample number obtained from MIL-STD-105 shall denote the number of test groups to be taken from the lot. In the event that the lot does not contain rolls of tape of the width required for test samples, the manufacturer shall supply tape of the required width for use as inspection samples. The additional tapes shall be from the same manufacturing lot or lots as the tape undergoing inspection.

* 4.4.3 Quality conformance tests. Quality conformance tests are those tests performed on tapes produced in accordance with the requirements of this specification and have been submitted for acceptance under a contract or order. Quality conformance tests shall consist of the tests listed in Table II.

* 4.5 Examination of the end item. The end item shall be examined in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQL's) set forth below. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105 shall be expressed in units of rolls for examinations under 4.5.2 through 4.5.5 inclusive, and in units of shipping containers for examination under 4.5.6, of the specified type and color of tape.

* 4.5.1 Inspection levels and acceptable quality levels (AQL's) for examinations. The inspection levels for determining the sample size and the acceptable quality levels (AQL's), expressed in defects per 100 units, shall be as specified in Table III.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

TABLE II. Instructions for Testing.*

| Characteristics | Specification reference | | Requirements applicable to | | Number of specimens per test determination, Minimum | Results reported as | |
|---|---|---|---|---|---|---|---|
| | Requirements | Test Method | Individual unit | Lot Average | | Pass or fail 1/ | Numerically or nearest 2/ |
| Thickness | Table I | 4.7.1 | X | -- | 5 | -- | 0.001 inch |
| Breaking strength | Table I | 4.7.2 | X | -- | 5 | -- | 0.1 lb per inch width |
| Adhesion to magnesium | Table I | 4.7.3.2.1 | X | -- | 5 | -- | 1.0 oz/ inch width 3/ |

NOTE: 1/ If failure is indicated, report either description or numerical point of failure, as applicable.

2/ Test reports shall include all values on which results are based.

3/ Failure of more than one specimen shall reject material.





Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

TABLE III. Inspection levels and AQL's.

| Examination Paragraph | Inspection levels | AQL |
|---|---|---|
| 4.5.2 | I | 4.0 |
| 4.5.3 | S-2 | 4.0 |
| 4.5.4 | S-2 | 4.0 |
| 4.5.5 | S-2 | Zero |
| 4.5.6 | S-2 | 4.0 |

NOTE: The same rolls of tape shall be used for examinations under 4.5.2 through 4.5.5 inclusive. The yardage used for examination under 4.5.2 and the rolls used for examinations under 4.5.4 and 4.5.5 shall be within the rolls randomly selected under 4.5.1.

* 4.5.2 Examination of the end item for defects in appearance and workmanship. The sample unit for this examination shall be 1 yard.

| EXAMINE | DEFECT |
|---|---|
| Appearance | Tape does not consist of a backing with a coating of adhesive on one side only. |
| | Dirty areas or spots. |
| | Holes, tear, cut, cracks, or sharp creases. |
| | Non-clean cut, broken, uneven or sticky edges. |
| Workmanship | Adhesive coating not evenly and smoothly applied over entire area of one side of backing; any bare spots or lumps. |

4.5.3 Examination of the end item for defects in construction. The sample unit for this examination shall be one roll.

| EXAMINE | DEFECT |
|---|---|
| Tape on rolls | Rolls not uniformly and smoothly wound; not wound on either paper-fiber or plastic core; core not same width as tape. |
| | Adhesive side not wound on inside of roll. |
| | Core crushed, broken, mutilated, or collapsed. |
| | Markings omitted, incorrect, incomplete, illegible, or not as specified. |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

| EXAMINE | DEFECT |
|---|---|
| Unwinding of rolls (examine both sides) | Does not unroll evenly and uniformly without ravelling. |
| | Backing breaks, delaminates, or splits during unwinding. |
| | Tape wound unevenly causing wrinkles or crease within the roll. |
| | When unwound, tape sticks together to the extent that unrolling causes tearing or injury to surface; adhesive separates from backing. |
| | Roll not continuous. |
| | More than 3 splices in roll; splices not evenly and nearly made; splice comes apart during unwinding. |

* 4.5.4 Examination of the end item for dimensional defects. The sample unit for this examination shall be one roll.

| EXAMINE | DEFECT |
|---|---|
| Width of tape | Varies by more than ±1/16 inch from width specified. |
| Cores | Inside diameter of core greater than 3-1/16 inches. |
| | Inside diameter of core less than 3 inches. |

* 4.5.5 Examination of the end item for length per roll. The sample unit for this examination shall be one roll. The average length shall be not less than the length specified in the procurement document.

* 4.5.6 Examination of the end product for defects in packaging, packing, and markings. An examination shall be made to determine that packaging, packing, and markings as required by Section 5 of this specification are complied with. The sample unit for this examination shall be one shipping container.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

| EXAMINE | DEFECT |
|---|---|
| Packaging: (as applicable) | Not level specified; not in accordance with contract requirements. |
| Individual roll (when specified) | Not individual-roll packaged and adequately secured as specified; inside or outside centering device omitted. |
| | Packaging material not as specified; closures not accomplished by specified or required methods or materials. |
| Multiple roll (when specified) | Not packaged as specified; not adequately secured in position; inside or outside centering device omitted; waxed paper (or equivalent) separators missing from between rolls. |
| Packing (as applicable) | Not level specified; not in accordance with contract requirements. |
| | Liner material (as applicable) omitted, damaged, or not as specified. |
| | Container material damaged or not as specified. |
| | Closures not accomplished by specified or required methods or materials. |
| Count | Number of rolls per container less than indicated. |
| Weight (interior and exterior) | Gross or net weight exceeds contract requirements. |
| Markings (interior and exterior) | Illegible, incorrect, incomplete, omitted or not in accordance with contract requirements. |
| | Markings not in accordance with 5.2. |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

* 4.6 Test conditions.

* 4.6.1 Sample rolls. The first three turns of tape shall be removed and discarded from each sample roll of tape prior to testing.

* 4.6.2 Tape conditioning. All tape selected for testing shall be conditioned for 24 hours at 73° ± 3.5°F (23°± 2.0°C) and 50 ± 5 percent relative humidity and shall be tested under these conditions unless otherwise specified in the method of tests.

* 4.6.3 Device for application of tape. The device for application of the tape for test purposes shall consist of a horizontally mounted, free turning, rubber coated roller with dimensions conforming to the following:

| | |
|---|---|
| Total weight | 4.5 ± 0.1 pounds |
| Width | 2.5 ± 0.1 inches |
| Diameter | 3.25 ± 0.1 inches |

The roller shall be faced with 1/4 inch of rubber with a shore durometer hardness of 75 ± 5. The rolling surface shall be void of any concave or convex deviations which might prevent the application of uniform pressure on the rolling surface. The roller shall be propelled lengthwise back and forth over the tape in such a manner that only the weight of the roller rests on the tape and at the rate of approximately 12 inches per minute. The device can be either manually or mechanically operated. The adhesive side of the tape shall not be touched during application. For referee purposes the device shall be mechanically operated.

* 4.6.4 Test specimens. Five specimens shall be run for each test unless otherwise specified. Three feet of tape shall be removed from the running end of each roll for the initial starting point. There shall be a 5-foot interval between specimens for each test.

* 4.7 Test methods.

* 4.7.1 Thickness. The thickness shall be determined in accordance with Method No. 173 of UU-P-31.

* 4.7.2 Breaking strength. The breaking strength shall be determined in accordance with Method No. 5102 of FED-STD-191.

* 4.7.3 Adhesion.

* 4.7.3.1 Preparation of test panel. A panel at least 2 inches wide by 4 inches long and 1/16 inch thick shall be prepared from magnesium conforming to composition AZ31B-H24 with chrome pickle treatment of MIL-M-3171, Type 1. The panel shall be thoroughly cleaned with mineral spirits



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

conforming to Type I, Grade A or Type II, Grade A of TT-T-291. After cleaning, a specimen of the tape 1 by 8 inches shall be applied to the prepared panel so that a 1 by 4 inch area of the panel is covered by the tape. The tape shall be applied by rolling the tape application device back and forth once in the lengthwise direction over the tape.

* 4.7.3.2 Procedures.

* 4.7.3.2.1 Adhesion to magnesium. The free end of the tape shall immediately be doubled back to 180 degrees, and one inch of the applied tape peeled off the panel surface at the folded end. The end of the magnesium panel from which the tape has been folded back shall be clamped in the lower jaw of a tensile testing machine, and the free end of the tape folded over and clamped in the upper jaw. The tensile machine shall be operated at a speed of 12 inches per minute. The pawls on the pendulum arm shall be taped back in an open position. The tensile tester shall be of such capacity that the test values will be read when the pendulum hangs between 9 degrees and 45 degrees from the vertical. After the first inch is removed from the panel during the test, the average tension required to remove the remainder of the tape from the panel, except for the final one inch, shall be determined. Each of the determinations shall be recorded. Failure of more than one specimen to meet the requirements listed in Table I shall be considered cause for rejection for the material represented.

* 4.7.3.2.2 Adhesion after accelerated aging. Place a 1-inch wide roll of tape in a dessicator containing a glycerine-water solution having a specific gravity of 1.103 at 75 ± 2 degrees F to effect an atmosphere of 85 ± 2 percent relative humidity. The roll shall be placed in a horizontal position on a perforated plate above the solution. The dessicator shall be sealed and maintained at a temperature of 100 ± 2 degrees F for 14 days. After the conditioning period, the roll shall be removed from the dessicator and examined for roll distortion, telescoping, and excessive adhesive flow on the sides. A specimen of the tape shall be applied on a magnesium panel prepared as in 4.7.3.1, and the tape subjected to the adhesion test of 4.7.3.2.1.

* 4.7.3.2.3 Adhesion after immersion. The tape shall be applied to the magnesium panel, prepared as in 4.7.3.1, prior to immersion. Using two separate containers, place in one di-2 ethylhexysebacate (see 6.4), and in the other a petroleum base oil conforming with Type VI of TT-S-735 (ASTM oil No. 3 - See 6.4). The panels shall be completely submerged in the fluids. The two containers shall then be placed in an oven maintained at 155 ± 2 degrees F for 48 hours. After this immersion period the panels shall be removed and blotted repeatedly to remove any excess fluid. The tapes shall then be subjected to the adhesion test of 4.7.3.2.1.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

* 4.7.4 Effect of paint.

* 4.7.4.1 Preparation of test panels. Two panels, 16 by 5 by 0.040 inch thick, fabricated from QQ-A-250/5 aluminum, shall be used. The panels shall be thoroughly cleaned with a solvent conforming to Type I, Grade A or Type II, Grade A of TT-T-291. Three pieces of 1-inch wide tape shall be placed lengthwise on each panel. The edge of the first strip shall be placed 1/2 inch from either edge of the panel. There shall then be a 1/2 inch interval between the location of the remaining two strips. The tape shall be applied, in the long direction, by finger pressure, and the excess overhang shall be trimmed flush to the edges.

* 4.7.4.2 Overcoating. Spray each of the two test panels with one coat of epoxy polyamide primer conforming to MIL-P-23377 to a dry film thickness of 0.6 to 0.8 mil and air dry for one hour. Then apply a coat of white polyurethane paint conforming to MIL-C-81733 to a total dry film thickness of 2.2 to 2.8 mils.

* 4.7.4.3 Procedure. Twenty-four hours after the panels have been overcoated they shall be examined for conformance to the requirements of Table I.

* 4.7.5 Corrosion.

* 4.7.5.1 Preparation of test panel. The cleated test panel assembly, illustrated in Figure 1, shall be fabricated and prepared as follows:

(a) One panel, 5 by 16 by 1/16-inch thick shall be fabricated from QQ-A-250/5 aluminum.

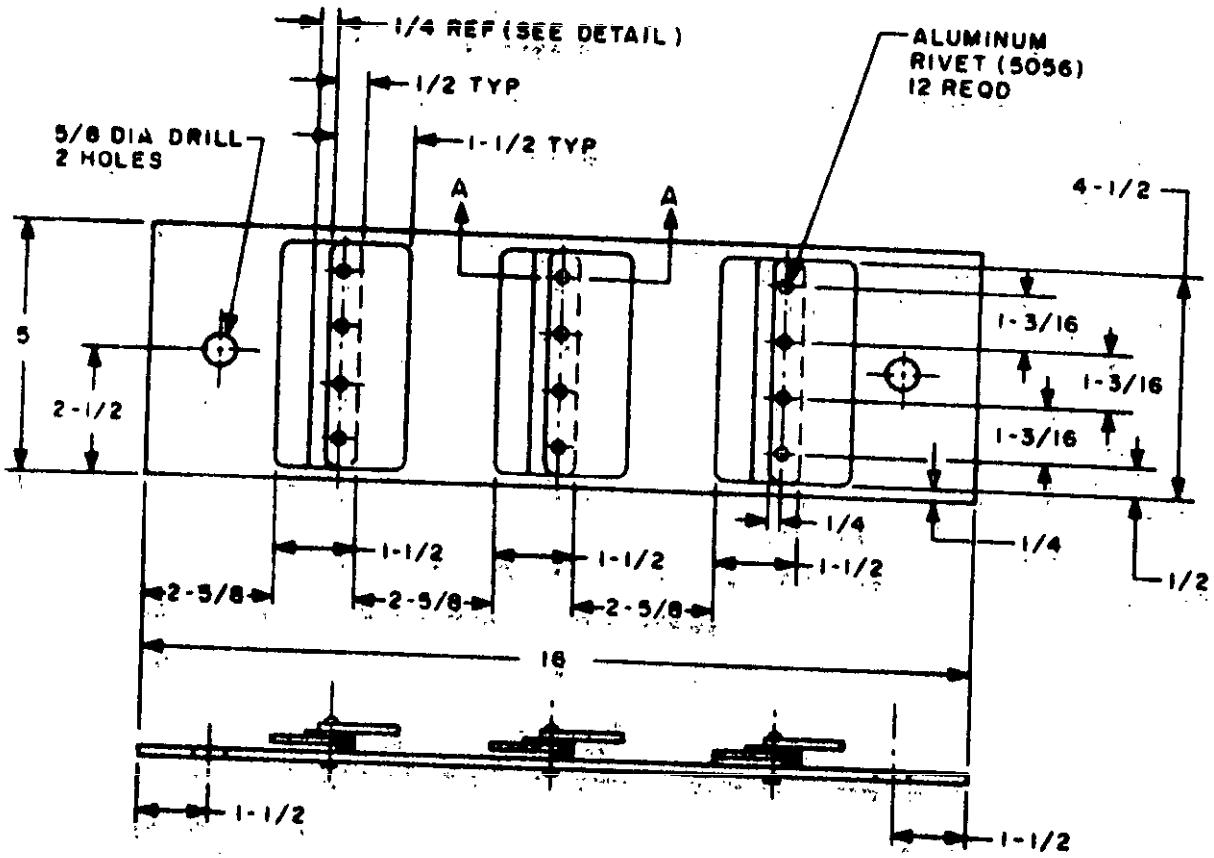
(b) Three pairs of cleats, each pair consisting of one 4-1/2 by 1-1/2 by 1/16-inch thick QQ-M-44 magnesium cleat with chrome pickle treatment and one 4-1/2 by 1-1/2 by 1/16-inch thick QQ-A-250/13 aluminum cleat.

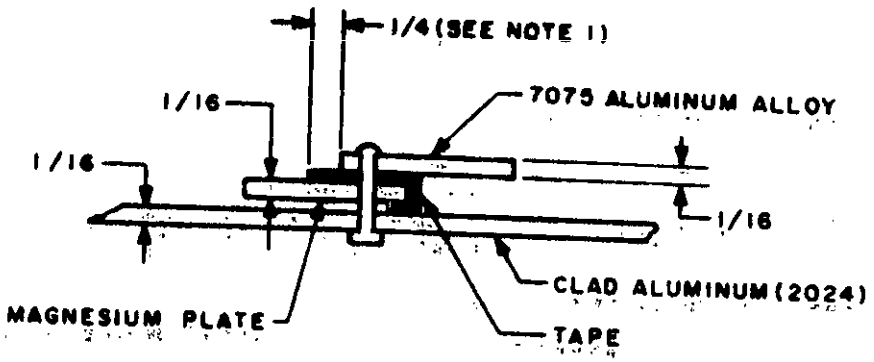
(c) Twelve 5056 aluminum, brazier head rivets, 1/8 inch in diameter and 3/8 inch long.

(d) The cleats shall be positioned on the panel as shown in Figure 1 with the magnesium cleat in contact with the panel and the aluminum cleat on top of the magnesium cleat. The cleats shall be clamped or otherwise secured to the panel and the cleats and panel shall then be marked to enable correct rivet hole re-alignment on final assembly. The rivet holes shall then be drilled at the spacing shown on Figure 1. Each rivet hole shall be drilled completely through both cleats and the panel simultaneously, but it shall not be necessary to drill all twelve holes at the same time.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.




SECTION A-A

NOTE:
1. TAPE MUST EXTEND 1/4 INCH BEYOND EDGE OF ALUMINUM CLEAT AND MUST BE APPLIED TO MAGNESIUM SURFACE.

DIMENSIONS IN INCHES.

FIGURE 1. Test panel assembly.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

(e) The panel shall be thoroughly cleaned with butyl or ethyl acetate and then coated with an epoxy polyamide primer conforming to MIL-P-23377 to a dry film thickness of 0.6 to 0.8 mil. Air dry for one hour, then apply a coat of white polyurethane paint conforming to MIL-C-81773 to a total dry film thickness of 2.2 to 2.8 mils. Air dry for 48 hours before testing.

(f) The cleats shall be thoroughly cleaned with mineral spirits conforming to Type I, Grade A or Type II, Grade B of TT-T-291.

(g) Tape shall be applied by finger pressure to the top of each magnesium cleat. The tape shall be applied, as illustrated in Figure 1, so that 1/4 inch of the tape will extend beyond the aluminum cleat on final assembly. The remaining tape shall be passed over the edge and applied to the under surface of the magnesium cleat as shown in Figure 1 to effect a complete separation of the magnesium and aluminum cleats.

(h) The cleats and panel shall be matched by means of the matching marks and riveted to form the final assembly illustrated in Figure 1.

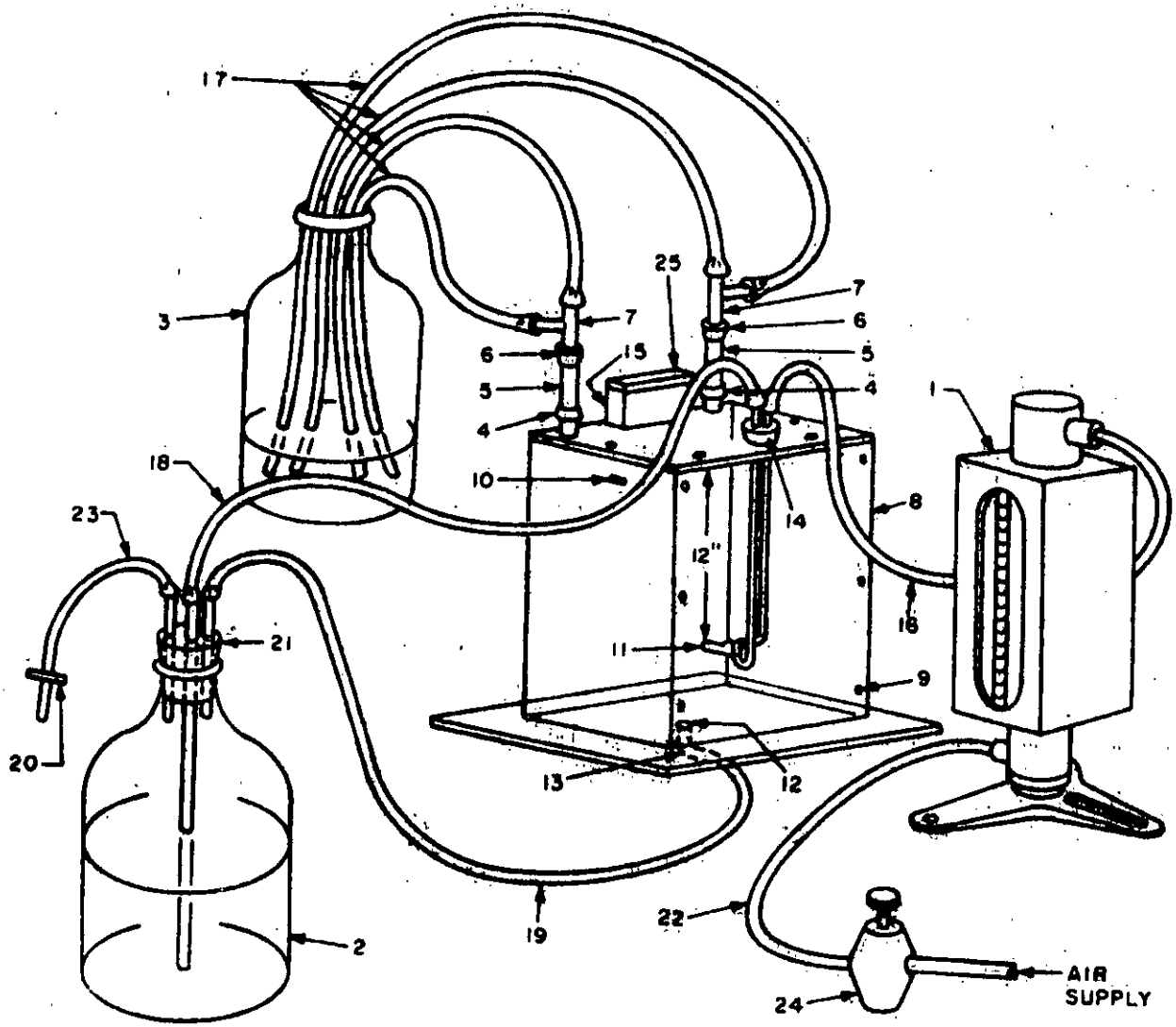
* 4.7.5.2 Procedure. The assembly shall be placed under an R.S. Sunlamp for 120 hours. The lamp shall be positioned 15 inches from the base of the assembly. A screen of opaque material shall encircle the exposure area. A constant line voltage to the lamp shall be maintained so that a direct scale reading of 32 units is obtained on a G.E. Sunlamp Tester No. 113 or its equivalent, used without shield.

The assembly shall be removed from under the lamp and positioned lengthwise in the specimen holder of the sea spray cabinet. Details of the cabinet construction are found in Figures 2, 3 and 4. Ten liters of the synthetic sea-water solution from the composition described in Table IV shall be placed in the supply jar. The box shall be secured and the air turned on to a flow rate (measured by a flowmeter) of 2.6 cubic feet per minute.

The assembly shall be exposed for 48 hours after which the rivets shall be removed. The tape shall then be removed, and the protected magnesium surface examined for evidence of pitting, corrosion, or etching.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.




1 - FLOWMETER 0-10 C.F.M.
2 - SALT WATER SUPPLY
3 - AIR SCRUBBER
4 - (2) RUBBER STOPPER
5 - (2) EXHAUST GLASS TUBES
6 - (2) RUBBER STOPPER
7 - (2) "T" TUBES
8 - SALT WATER CHAMBER
9 - SCREWS - 16 REQD
10 - PANEL HOLDER
11 - SPRAY NOZZLE
12 - DRAIN
13 - PLEXIGLASS TUBE
14 - RUBBER STOPPER
15 - PLEXIGLASS RISER
16 - AIR PRESSURE HOSE (SAFETY WIRED)
17 - EXHAUST HOSE
18 - SALT WATER SUPPLY HOSE
19 - SALT WATER RETURN HOSE
20 - SMALL "C" CLAMP
21 - RUBBER STOPPER
22 - AIR SUPPLY HOSE (SAFETY WIRED)
23 - BREATHER HOSE
24 - AIR PRESSURE REGULATOR
25 - PRESSURE SENSITIVE TAPE

FIGURE 2. Sea spray apparatus.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

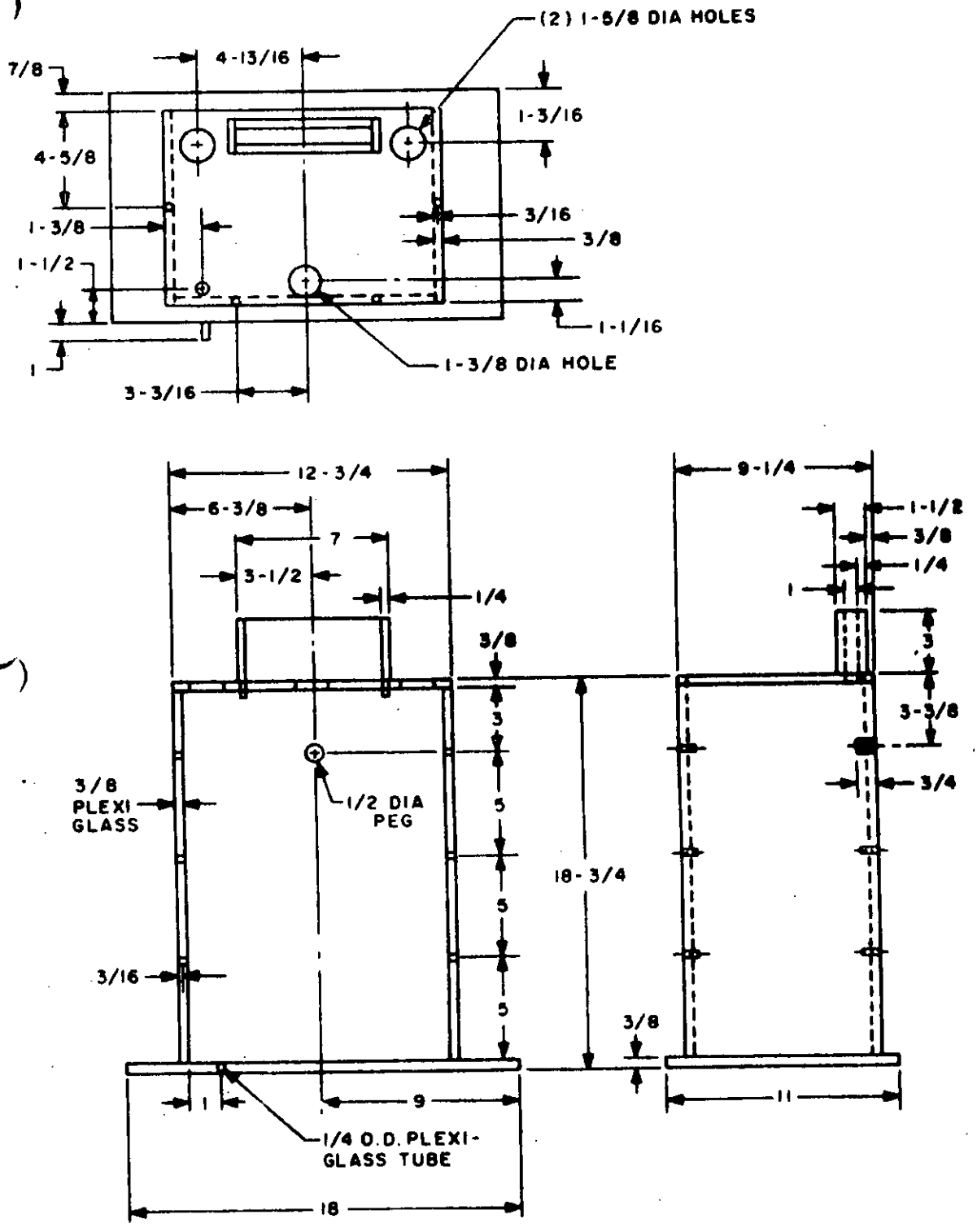
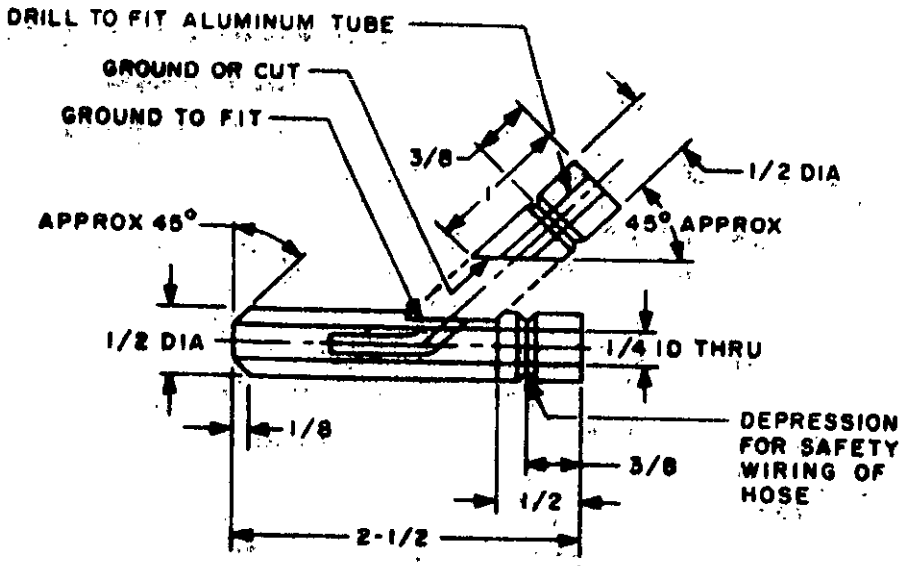


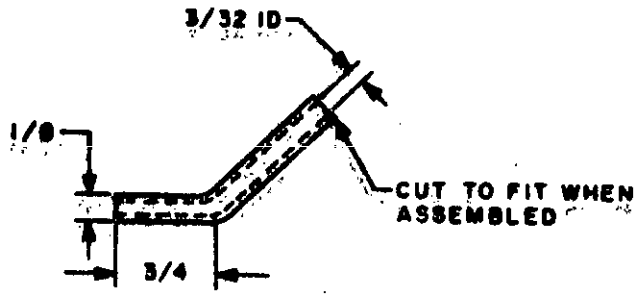

FIGURE 3. Sea spray chamber.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.




PLEXIGLASS RODS




GLASS TUBE

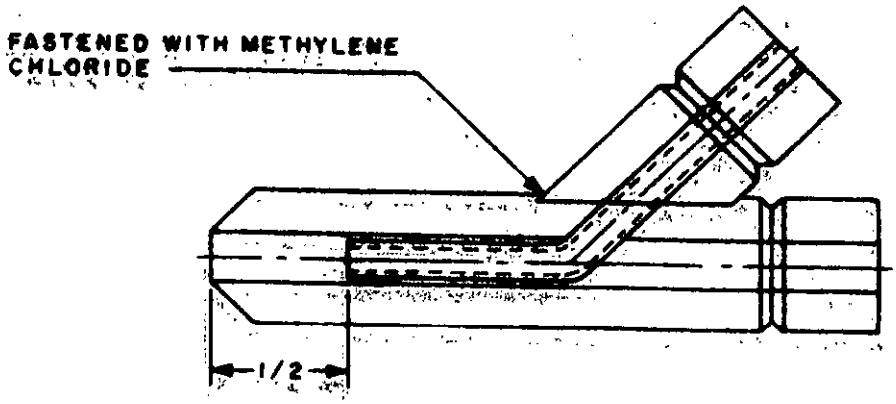



FIGURE 1. Nozzle for sea spray chamber.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

TABLE IV. Composition of synthetic seawater.

| Constituent | Grade or specification | Percentage by weight | Grams per liter of distilled water |
|---|---|---|---|
| Sodium chloride (NaCl) | Technical | 5.0 | 50.0 |
| Magnesium chloride ($MgCl_2$ $6H_2O$) | Technical | 2.2 | 22.0 |
| Calcium chloride ($CaCl_2$ $2H_2O$) | O-C-105 | 0.32 | 3.2 |
| Sodium sulfate ($Na_2SO_4$) | Technical | 0.8 | 8.0 |

* 4.7.6 Weathering (accelerated outdoor exposure). The contractor shall make available to the government representative a certificate of compliance (see 6.6) with the weathering (accelerated outdoor exposure) test, performed as follows: Two corrosion test assemblies shall be prepared as specified in 4.7.5.1 but modified by a Dow No. 7 surface treatment of the magnesium cleat, in accordance with Type III of MIL-M-3171. The assemblies shall be exposed for 120,000 ± 5 percent Langleys in an accelerated outdoor weathering machine (see 6.7). One panel shall be exposed to an equitorial mount with mirrors for acceleration with water (EMMA) and one panel to an equitorial mount with mirrors for acceleration with water (EMMAQUA). After exposure the assemblies shall be removed from the mounts and examined for conformance to the requirements of weathering (accelerated outdoor exposure) in Table I.

* 4.7.7 Accelerated aging. The roll of tape after being conditioned as in 4.7.3.2.2 shall be visually examined for roll distortion, telescoping and excessive adhesive flow on side of roll.

* 4.7.8 Shelf storage. The contractor shall make available to the government representative a certificate of compliance (see 6.6) with the shelf storage test performed as follows: a one inch wide roll of tape in the container shall be placed unopened in a conditioned room, as specified in 4.6.2, for a period of one year. At the end of the storage period, the roll of tape shall be examined for roll distortion, telescoping and excessive adhesive flow on the sides of the roll. The tape shall then be subjected to the adhesion to magnesium test 4.7.3.2.1.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

## 5. PREPARATION FOR DELIVERY

* 5.1 Packaging and packing. Rolls of tape shall be packaged Level A or commercial, packed Level A, B, or commercial, as specified in the contract or order, in accordance with PPP-T-680.

5.2 Marking. In addition to the marking specified in PPP-T-680, unit and exterior containers shall be durably and legibly marked with the following information:

Federal Stock No.
Width
Length of Tape
Contract or Order
Date of manufacture (month and year)
Manufacturer's name and address

## 6. NOTES

6.1 Intended use. The pressure-sensitive tape covered by this specification is intended for use as a separator to prevent galvanic attack between dissimilar metals.

6.2 Ordering data. Requests, requisitions, schedules, contracts or orders should contain the following:

(a) Level of packaging (5.1).
(b) Level of packing (5.1).
(c) Whether single or bulk packaging is required.
(d) Width if other than 1 inch (3.3.2).
(e) Length of tape in roll if other than 72 yards (3.3.2).

6.3 First article inspection provisions. Contracts or orders shall specify the following provisions for the first article inspection:

* 6.3.1 Whether first article inspection is required. When a contractor is in continuous production of tapes from contract to contract, consideration should be given to waive the first article inspections. If inspection is required, indicate:

(a) Where the first article inspection is to be conducted (at the contractor's plant or Government or commercial laboratory).

(b) That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

6.4 Source of material. The material called out in the adhesion after immersion test may be obtained from the following source:

| Material | Source |
|---|---|
| Di-2 ethylhexysebacate (Plexol 201) | Rohm and Haas<br>Washington Square<br>Philadelphia, PA 19105 |
| Petroleum base oil (ASTM Oil No. 3) | Sun Oil Co.<br>Special Products Division<br>1616 Walnut Street<br>Philadelphia, PA. 19103 |

* 6.5 Storage. The tape should be stored in the original container, preferably in a cool location. It should not be stored in close proximity to steam pipes, radiators, or other sources of heat.

* 6.6 Certificate of compliance. The certificate of compliance shall cite the data or the location and availability of data, i.e., the quality control upon which the certificate is based.

6.7 Location of accelerated outdoor exposure facilities. Accelerated outdoor exposure (EMMA and EMMAQUA) facilities are available through Desert Sunshine Exposure Testing, Box 185, Black Canyon Stage, Phoenix, Arizona 85020.

* 6.8 Changes from previous issue. The outside margins of this specification have been marked "*" to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationships to the last previous issue.

Custodian:
Navy - AS
Army - GL
Air Force - 99

Preparing activity:
Navy - AS
Project No. 7510-0300

Review activity:
Army - MI
Air Force - 69

User activities:
Army - ME, GL



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

**INSTRUCTIONS:** In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide [illegible]stions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and [illegible]d. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was so rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STAT

[redacted]

[illegible]FICIAL BUSINESS
[illegible]NALTY FOR PRIVATE USE $300

**BUSINESS REPLY MAIL**

FIRST CLA[illegible] PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

ENGINEERING SPECIFICATIONS AND STANDARDS DEPARTMENT
(CODE 93)
NAVAL AIR ENGINEERING CENTER
LAKEHURST, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |
| | |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil - Downloaded: 2026-10-19T15:41Z
Check the source to verify that this is the current version before use.